

Steering ESM2-650M: What Moves, What Does Not, and How to Tell in Advance

Dmitriy Ivkov

August 21, 2026

Abstract

We tried to steer nine sequence properties in the protein language model ESM2-650M using activation steering, under one fixed method and one acceptance protocol fixed in advance. One number, measured before any generation—the amino-acid *compositional separation* between the low- and high-label groups the steering vector is built from—predicts both whether a property is steerable and how large the effect is. The proxy’s correlation with the true label does not: the best-validated proxy in the whole set ($r = 0.80$, binding) steered nothing, while the weakest runnable one ($r = 0.22$, catalytic turnover) passed on all three seeds. Effect size grows with separation, from a flat null to 72% of the natural property gap. Separation is necessary but not sufficient—two well-separated targets still failed, one because the model ignores the direction and one because the property is *positional*. That last case is a structural limit: a vector added uniformly at every position can change how much of a residue a sequence contains but not where it goes, so this method can move aggregate compositional properties (transmembrane fraction, catalytic composition, charge, glycosylation density) but cannot install positionally-localized features (signal peptides, active-site geometry, domain boundaries). All results are computational.

1 How we obtained the results

The steering method (identical for every target). The model is ESM2-650M [Lin et al., 2023]. To generate, we mask 30% of a sequence’s positions and fill them in a single argmax pass (`mask_fill_generate`). To steer, we build a direction as the per-layer difference between the mean activations of 150 high-label and 150 low-label proteins, and add it to all 33 layers, the same amount at every sequence position, scaled by a strength α . This is the standard difference-of-means activation steering used for protein language models [Huang et al., 2025]. As a control we use a random direction with the same norm. We evaluate on 150 held-out proteins, score each generation with a property-specific compositional proxy, and report the effect as a paired bootstrap contrast (10,000 resamples) between the real and random directions. The masked-fill decoder collapses into repetitive low-complexity output at $\alpha \geq 0.35$, so we sweep dose only on the valid window $\{0.10, 0.15, 0.20, 0.25\}$; this caps the achievable effect but keeps every comparison inside a regime where the outputs are real sequences.

The acceptance protocol (fixed before the runs). To keep a flattering number from being read as a capability, each target had to clear six criteria set in advance:

1. beat the random-direction control with a real bootstrap confidence interval;
2. show a coherent dose-response over at least three strengths;

3. survive **residue exclusion**—stay significant after the two most frequently substituted residues are removed from the score (every proxy is a function of composition, so steering can cheat by just enriching favored residues);
4. use a proxy validated against real experimental labels *before* the run ($|r| \geq 0.15$, checked in code before the model even loads);
5. beat any prior technique for the property;
6. be adequately powered ($n \geq 150$).

Failing 1–4 outright is a KILL; passing most but failing a dose check or residue exclusion is AMBIGUOUS; clearing all operative criteria is a PASS. We reran every candidate on three independent seeds.

The separation gate. Before spending any GPU time, we measure the mean per-residue amino-acid composition L_2 distance between the low- and high-label groups that build the vector. It began as a screen for one early null and turned out to be the single most predictive quantity in the whole effort.

Targets, data, and proxies. Nine properties, chosen to span a range of separation and to contrast properties intrinsic to a single sequence against relational ones. Transmembrane fraction, DNA-binding, glycosylation density, disulfide content, signal peptide, zinc-finger and calcium-binding domains came from reviewed UniProt [The UniProt Consortium, 2023]; cross-protein k_{cat} from DLKcat [Li et al., 2022]; single-backbone binding from ProteinGym [Notin et al., 2023]. Proxies use published biophysical scales: mean Kyte–Doolittle hydropathy for membrane character [Kyte and Doolittle, 1982], Lys+Arg fraction for nucleic-acid interfaces [Luscombe et al., 2001], a glycine-minus-arginine contrast for the turnover–stability tradeoff [Siddiqui and Cavicchioli, 2006]. Each proxy was validated against the real label on a held-out split before its run.

2 Results

2.1 Separation predicts steerability and effect; proxy correlation does not

Table 1 lists all nine targets. Sorting by outcome, effect size follows the pre-run separation and ignores the proxy’s label correlation. The two extremes make the point. Single-backbone binding has the best-validated proxy anywhere in the project (held-out $r = 0.80$, and it generalizes to unseen positions), yet it steers *nothing* at any strength—because its low- and high-label sequences are one- and two-residue mutants of the same 188-residue backbone, so they are nearly identical in composition (separation 0.003) and the difference-of-means vector has almost no signal to carry. Catalytic turnover has the weakest runnable proxy ($r = 0.22$) but passes on all three seeds. Proxy strength and steerability are not even weakly related.

The passing targets calibrate the relationship: separation 0.003 $\rightarrow$ null, 0.039 $\rightarrow$ a quarter of the natural binder gap, 0.086 $\rightarrow$ nearly three-quarters of the way to the membrane-protein distribution (proxy $-0.363 \rightarrow +0.266$ at $\alpha = 0.25$ against a natural gap of 0.871). The binding null was originally read as “binding is not steerable.” Rebuilding the same target from *cross-protein* data—DNA-binding, where a protein binds the backbone by its own sequence—lifts separation about $12\times$ (0.003 $\rightarrow$ 0.039) and turns the null into a significant, residue-robust, dose-responsive effect. So the null was an artifact of single-backbone data, not a property of binding, and we predicted the recovery from separation before running it.

Table 1: Nine steering targets on ESM2-650M, one fixed method, sorted by outcome. *Sep.* is the pre-run compositional distance between the vector groups; *proxy r* is the held-out correlation of the scoring rule with the real label; *effect* is, where a natural low→high proxy gap is defined, the fraction of that gap reached at $\alpha = 0.25$. Effect tracks separation, not proxy *r*.

Target	Sep.	Proxy <i>r</i>	Outcome	Effect / why it failed
Binding, single-backbone	0.003	0.80	KILL	flat null — no separation
Zinc-finger domain	0.028	0.29	KILL	direction anti-correlated
Calcium-binding (EF-hand)	0.038	0.16	KILL	model ignores the direction
Signal peptide	0.035	0.74	KILL	positional, not compositional
Disulfide (Cys fraction)	0.026	0.17	pass (2/3)	tiny (~ 0.002)
Glycosylation density	0.039	0.41	pass (3/3)	tiny (~ 0.003)
DNA-binding propensity	0.039	0.25	pass (2/3)	25% of natural gap
Catalytic turnover (k_{cat})	0.02–0.05	0.22	pass (3/3)	moderate
Transmembrane fraction	0.086	0.79	pass (3/3)	72% of natural gap

2.2 Separation is necessary but not sufficient

Screening by separation kills the null mode but does not guarantee success. Two targets with separation in the passing band still failed, for different reasons. Calcium-binding (separation 0.038) produces no significant steering at any safe strength—a well-separated group whose direction the model’s residual stream simply does not act on. Zinc-finger’s cysteine/histidine direction turns *anti-correlated* with its own proxy after steering. High separation is a prerequisite; the model must also respond to the direction, and the direction must encode the property rather than a correlate.

2.3 The method steers composition, not position

The sharpest failure is the signal peptide. Its proxy—mean hydropathy over the first 30 residues—is the best-validated candidate in the whole set (held-out $r = 0.742$), and it gives a clean dose-response (N-terminal hydropathy $0.059 \rightarrow 0.100 \rightarrow 0.152 \rightarrow 0.251$). It still KILLS on residue exclusion: remove the two dominant substituted residues and the effect collapses to non-significant. The vector raises N-terminal hydropathy only by dumping bulk hydrophobic residues across the *entire* sequence; it cannot *concentrate* them at the N-terminus, which is what makes a signal peptide. Glycosylation density passes for the mirror-image reason: its proxy is a sequon count summed over the whole sequence, a genuinely aggregate quantity a mean-pooled vector can nudge.

The general statement: a direction that is averaged over positions and added identically to every position can change *how much* of a residue type a sequence has, but not *where* it sits. Aggregate compositional properties—transmembrane fraction, catalytic composition, DNA-binding charge, glycosylation density—are reachable. Positionally-localized features—signal peptides, and by the same argument active-site geometry, domain boundaries, termini motifs—are not, for any proxy, until the injection itself is made position-aware.

2.4 What the acceptance protocol caught elsewhere

The predictor is only trustworthy because the same protocol threw out flattering numbers in the studies that fed it, and each rejection changed a conclusion:

- An immune-endpoint proxy correlated $r = 0.427$ with an allele-aggregated binding label but only 0.100 with measured T-cell response, and flipped from +0.379 to -0.323 when validation folds were grouped by source organism — it measured the wrong endpoint, and was stopped before generation.
- A layer-subset thermostability run looked competitive only because the all-layer arm’s outputs had collapsed under the low-complexity filter; inside the shared valid range the subset kept just 43–59% of the full effect.
- A solubility contrast of +0.0125 fell to +0.00035 (interval crossing zero) once its two dominant substituted residues were excluded.
- A disorder effect repeated across three runs but survived residue exclusion in only two of them.

These are the same guards—endpoint validity, evaluable denominator, composition dependence, whole-run stability—that make a passing row in Table 1 mean something.

3 Takeaways and limitations

For this method, steerability and its ceiling are predictable before any compute from the compositional separation of the vector-building groups, and an entire class of properties (positional features) is ruled out on structural grounds, not by trying and failing one at a time. The next moves are to change the intervention—position-aware injection—rather than the target list, and to check the strongest compositional gain (transmembrane) against a real assay.

The limits are firm. Every endpoint is computational: no wet-lab or structural assay shows that any generated sequence actually has more thermostability, turnover, membrane character, binding, or glycosylation; every “effect” is a change in a compositional proxy. Everything uses one model, one decoder, and one injection geometry, so the composition-only limit is specific to that geometry. The usable dose range is narrow because the decoder degenerates above $\alpha = 0.25$. The proxies are compositional and non-mechanistic—a pass shows the model can be pushed toward an amino-acid signature that correlates with the property, not that structure or function is preserved. The separation predictor is calibrated on nine targets from one model; whether the same ordering holds for other models or injection schemes is untested.

References

- Long-Kai Huang, Rongyi Zhu, Bing He, and Jianhua Yao. Steering protein language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 26247–26260. PMLR, 2025. URL <https://proceedings.mlr.press/v267/huang25x.html>.
- Jack Kyte and Russell F. Doolittle. A simple method for displaying the hydropathic character of a protein. *Journal of Molecular Biology*, 157(1):105–132, 1982. doi: 10.1016/0022-2836(82)90515-0.
- Feiran Li, Le Yuan, Hongzhong Lu, Gang Li, Yu Chen, Martin K. M. Engqvist, Eduard J. Kerkhoven, and Jens Nielsen. Deep learning-based kcat prediction enables improved enzyme-constrained model reconstruction. *Nature Catalysis*, 5(8):662–672, 2022. doi: 10.1038/s41929-022-00798-z.
- Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom

- Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.
- Nicholas M. Luscombe, Roman A. Laskowski, and Janet M. Thornton. Amino acid–base interactions: a three-dimensional analysis of protein–DNA interactions at an atomic level. *Nucleic Acids Research*, 29(13):2860–2874, 2001. doi: 10.1093/nar/29.13.2860.
- Pascal Notin, Aaron W. Kollasch, Daniel Ritter, Lood van Niekerk, Steffanie Paul, Hansen Spinner, Nathan Rollins, Ada Shaw, Ruben Orenbuch, Rose Weitzman, Jonathan Frazer, Mafalda Dias, Dinko Franceschi, Yarin Gal, and Debora S. Marks. ProteinGym: Large-scale benchmarks for protein fitness prediction and design. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023.
- Khawar Sohail Siddiqui and Ricardo Cavicchioli. Cold-adapted enzymes. *Annual Review of Biochemistry*, 75:403–433, 2006. doi: 10.1146/annurev.biochem.75.103004.142723.
- The UniProt Consortium. UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Research*, 51(D1):D523–D531, 2023. doi: 10.1093/nar/gkac1052.